

Vietnamese GraphRAG

Multi-hop Question Answering over Wikipedia

Capstone Review 1 — FPT University SE+AI

Huynh Quoc Trung (QE180038) Nhu Quang Anh (QE180005) Pham Ngo Dinh Khoi
(QE180110) Truong Dinh Thien (QE180018)

FPT University

Supervisors: Lê Trung Hiếu Đặng Nguyên Bình

Week 6 — June 2026

Table of Contents

- | | |
|-----------------------------------|----------------------------|
| ① Problem & Objectives | ⑦ Dataset |
| ② Users & Practical Applicability | ⑧ Related Works (3 papers) |
| ③ Innovation & New Technology | ⑨ AI Pipeline |
| ④ System Architecture | ⑩ Evaluation Metrics |
| ⑤ Requirements & SRS | ⑪ Team Assignment |
| ⑥ AI: Problem Framing | ⑫ System Scale & Timeline |

1. Problem Statement

Core Challenge

Complex questions about Vietnamese history, geography, and people often **span multiple Wikipedia articles** — impossible to answer with a single search.

User	Problem
Students / researchers	Must read 5–10 articles to synthesize an answer
Standard RAG systems	Only retrieves the nearest chunk — loses multi-hop links
Vanilla LLMs	Hallucinate Vietnamese facts, no verifiable citations

Research Gap

No publicly available Vietnamese GraphRAG system supports **multi-hop reasoning** over Wikipedia with **explainable citations**.

Project Goal

Build a fully local Vietnamese KGQA system that answers multi-hop questions with validated citations, with no dependency on external APIs.

2. Users & Practical Applicability

Real User Groups

- **FPT / HUST / UIT students** — academic knowledge lookup in Vietnamese
- **Researchers / journalists** — synthesize from Vietnamese Wikipedia sources
- **AI developers** — integrate into their own Vietnamese RAG pipelines

Applications

- **Vietnamese Knowledge Search** — natural language QA over local Wikipedia corpus
- **GraphRAG Benchmark Platform** — reproducible multi-hop evaluation for Vietnamese NLP

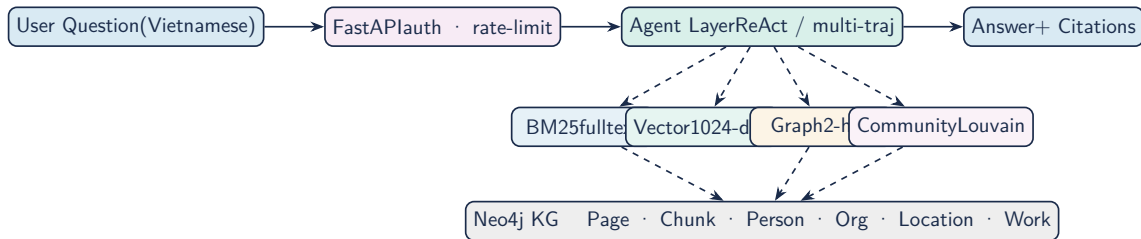
Expected Benefits

- Reduce manual navigation across Wikipedia articles
- Improve answer reliability through citations
- Enable multi-hop reasoning over Vietnamese knowledge

3. Innovation & New Technology

Innovation	Description
WRRF Hybrid Retrieval	BM25 + Vector + Graph + Community — no Vietnamese QA system uses all 4 signals
Multi-trajectory Agent	N parallel trajectories + majority voting (Thompson et al. arXiv:2506.19967)
Local-first sovereignty	Entire pipeline runs on local GPU; no data sent externally
ViWiki-MHR Dataset	First Vietnamese multi-hop dataset: KG walk + LLM rewrite + 5-layer QC
Typed Vietnamese KG	Neo4j, 4 entity types, 6 relation types from Vietnamese Wikipedia 2026

4. System Architecture



5. Requirements (SRS Summary)

Functional Requirements (Must Have)

ID	Requirement	Verifiable By
FR-01	Ingest Vietnamese Wikipedia articles	Unit test
FR-02	Answer single-hop factual questions	EM/F1 on ViQuAD2
FR-03	Answer multi-hop questions + citations	Hit Rate on ViWiki-MHR
FR-04	Bulk ingest from HuggingFace	Load test 10K articles
FR-05	Generate MHQA evaluation dataset	QC pipeline pass rate

Non-Functional Requirements

ID	Metric	Target	Achieved
NFR-01	P50 latency	<3s / <8s	51ms retrieval ✓
NFR-02	Context Hit Rate	>70%	64.67% ~
NFR-03	Test coverage	≥75%	85% ✓

6. AI: Problem Framing

Layer	Content
Business Problem	Complex Vietnamese questions need synthesis across multiple Wikipedia articles
AI Task	Multi-hop KGQA: Graph-augmented RAG with typed knowledge graph
Formal Definition	$Q \rightarrow \text{retrieve } \{P_1 \dots P_n\} \text{ from KG } G \rightarrow \text{generate } A + \text{ citations}$

Question Taxonomy

Type	Hops	Example
Simple (lookup)	1-hop	"Born in which year?"
Bridge	2-hop	"Where did the founder of [X] study?"
Comparison	2+ hop	"Who is older: A or B?"
Fan-out	3+ hop	"List all works by [X]"

Feasibility: 15 weeks, team of 4 — core pipeline already operational ✓

7. Dataset

Dataset	Size	Source	Purpose
Vietnamese Wikipedia	~590K articles	wikimedia/wikipedia	KG construction
ViWiki-MHR	~200 pairs	KG walk + manual	Multi-hop eval
UIT-ViQuAD 2.0	36,457 pairs	UIT-NLP (VLSP 2021)	Benchmark

Data Quality Pipeline

- **Stub filtering:** ≥ 200 chars (1.29M $\rightarrow$ ~590K articles)
- **Quality filtering:** redirect, disambiguous, single character articles, dates, lists of articles (index)
- **NFC Unicode normalization** for Vietnamese diacritics
- **5-layer QC:** well-formedness $\rightarrow$ grounding $\rightarrow$ dedup $\rightarrow$ answerability $\rightarrow$ schema

8. Related Works — Paper 1

Microsoft GraphRAG — Edge et al., 2024 (arXiv:2404.16130)

Hierarchical community detection (Leiden) → pre-generate community summaries → map-reduce at query time. **Original result:** 70–80% win rate vs. naive RAG on broad queries.

Applied in this project:

- Louvain community detection → assign chunks to communities
- Community summaries → 4th WRRF signal (weight = 0.15)
- Improves recall for broad thematic queries about Vietnamese history/geography

8. Related Works — Paper 2

Inference-Scaled GraphRAG — Thompson et al., 2025 (arXiv:2506.19967)

Sample N graph traversal trajectories with temperature scaling → aggregate via majority voting. **Original result:** Substantial gains over traditional GraphRAG on GRBench; 31.44% vs. 15.26% on hard multi-hop questions.

Applied in this project:

- AGENT_N_TRAJECTORIES config — N agents run in parallel with temperature scaling
- Majority voting → consensus answer
- Particularly effective for comparison and fan-out question types

8. Related Works — Paper 3

UIT-ViQuAD 2.0 — Nguyen et al., 2020 (arXiv:2009.14725)

36,457 QA pairs from Vietnamese Wikipedia. Answerable + unanswerable.

Dataset curation: 5,000 paragraphs from 174 Wikipedia articles; questions manually created per paragraph.

SotA baseline (VLSP 2021): 77.24% F1 / 67.43% EM on private test set.

Applied in this project:

- Primary benchmark for retrieval quality evaluation
- `src/viquad_adapter.py` adapts HuggingFace → eval pipeline

Research Gap

Single-hop only, no KG walk ⇒ need a multi-hop dataset ⇒ **Our ViWikiHop dataset**

9. AI Pipeline

Ingestion

Wikipedia $\rightarrow$ normalize $\rightarrow$ chunk(500/50) $\rightarrow$ NER
 $\rightarrow$ embed $\rightarrow$ Neo4j

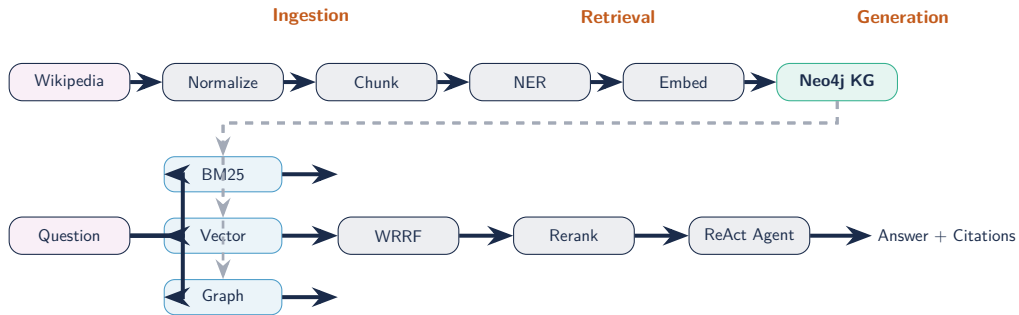
Retrieval (WRRF Fusion)

BM25 ($w=0.4$), Vector ($w=0.4$), Graph ($w=0.2$),
Community ($w=0.15$)
 $\rightarrow$ WRRF($k=60$) $\rightarrow$ rerank $\rightarrow$ top- K

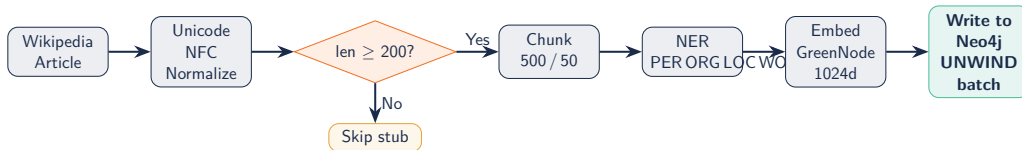
Answer Generation

Q + complexity detection
 $\rightarrow$ *Simple*: ReAct (≤ 6 iter, 6 tools)
 $\rightarrow$ *Complex*:
decompose sub-questions
 N trajectories (parallel)
majority vote
 $\rightarrow A$ + citations

9. AI Pipeline — End-to-End Overview

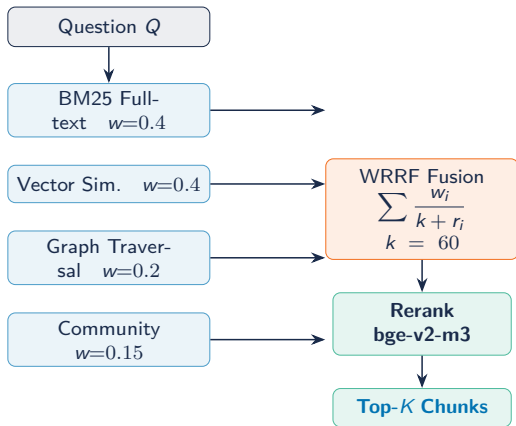


9a. AI Pipeline — Stage 1: Data Ingestion



Stub filter	NER backend	Embedding
1.29M → ~590K articles	wikilink (Typed F1 = 46.9%)	GreenNode 1024-dim

9b. AI Pipeline — Stage 2: WRRF Retrieval



Fusion Formula

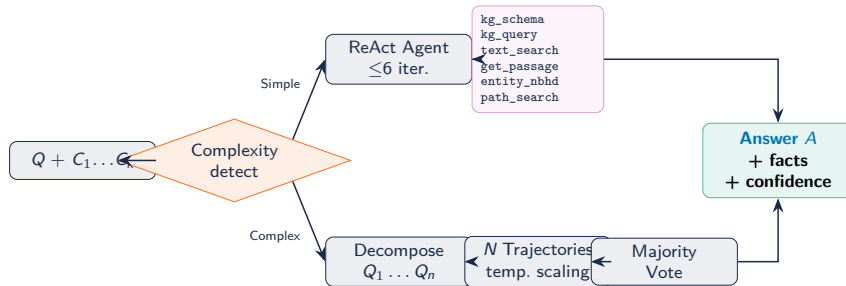
$$\text{score}(d) = \sum_i \frac{w_i}{k + \text{rank}_i(d)}$$

Signal	Weight
BM25 Fulltext	0.40
Vector (cosine)	0.40
Graph 2-hop	0.20
Community	0.15

$k = 60$ smooths rank-1 dominance.

Final reranker: BAAI/bge-reranker-v2-m3 (cross-encoder).

9c. AI Pipeline — Stage 3: Answer Generation



10. Evaluation Metrics

Exact Match (EM)

Binary: 1 if predicted answer matches any gold answer exactly (after normalization), 0 otherwise.

$$EM = \frac{1}{N} \sum_{i=1}^N \mathbf{1}[\hat{a}_i = a_i]$$

Normalization: lowercase, strip punctuation & articles.

Token F1

Soft overlap between predicted and gold answer token bags.

$$F_1 = \frac{2 \cdot P \cdot R}{P + R}, \quad P = \frac{|\hat{T} \cap T|}{|\hat{T}|}, \quad R = \frac{|\hat{T} \cap T|}{|T|}$$

Benchmark: UIT-ViQuAD 2.0

- 36,457 QA pairs from Vietnamese Wikipedia
- Answerable + unanswerable questions
- Standard MRC evaluation protocol (SQuAD-style)

SotA Reference (VLSP 2021)

Model	EM	F1
XLM-R (fine-tuned)	67.43%	77.24%

Our system targets comparable F1 after full corpus ingestion (~590K articles).

11. Team Assignment

Member	ID	Responsibility
Huynh Quoc Trung	QE180038	KG Construction · NER (6 backends) · Entity Resolution
Nhu Quang Anh	QE180005	WRRF Retrieval · Reranker · Evaluation Pipeline
Pham Ngo Dinh Khoi	QE180110	ReAct Agent · Local LLM · MHQA Dataset Gen
Truong Dinh Thien	QE180018	FastAPI · Job Management · Gradio Demo · Deploy

Dependency Chain

NER (Trung) → Entity quality → Graph signal (Anh) → Agent tools (Khoi) → FastAPI (Thien)

12. System Scale & Timeline

Use Case Points

UCP = 96.5 (UUCP = 134 · TCF = 0.9 · ECF = 0.8) → Feasible in 15 weeks ✓

Week	Milestone	Status
1–4	Core pipeline + SRS + Architecture	Done ✓
5–7	Full ingestion + Ablation + MHQA	In progress ~
8–10	QLoRA Text2Cypher + DPO alignment	Pending
11–13	WRRF tuning + final evaluation	Pending
14–15	Final report + demo + defense	Pending

Thank You

Questions & Discussion

Vietnamese GraphRAG — Multi-hop QA over Vietnamese Wikipedia

Local-first · Multi-hop · Citation-grounded · Fully local GPU

Appendix: Graph Schema

Node & Edge Types

`(:Page) -[:HAS_CHUNK] → (:Chunk)`
`(:Chunk) -[:MENTIONS_PERSON] → (:Person)`
`(:Chunk) -[:MENTIONS_ORG] → (:Organization)`
`(:Chunk) -[:MENTIONS_LOC] → (:Location)`
`(:Chunk) -[:MENTIONS_WORK] → (:Work)`
`(:Page) -[:LINKS_TO] → (:Page)`

Indexes

- Fulltext index on `chunk.text`
- Vector index on `chunk.embedding` (1024-dim)
- B-tree index on `entity.name`

Current scale:

998 pages · 19K chunks · 3.3K entities

Appendix: WRRF Formula

Weighted Reciprocal Rank Fusion

$$\text{score}(d) = \sum_i \frac{w_i}{k + \text{rank}_i(d)}$$

$$k = 60 \quad w_{\text{BM25}} = 0.4 \quad w_{\text{Vector}} = 0.4 \quad w_{\text{Graph}} = 0.2 \quad w_{\text{Community}} = 0.15$$

Cross-encoder reranker: BAAI/bge-reranker-v2-m3

Rescores top-20 candidates → returns top-5 to the agent

Appendix: NER Backend Comparison

Backend	Method	Best Use
Simple	Regex + keyword	Fast, no GPU
Underthesea	BIO tagging	Offline, general
PhoNLP	VnCoreNLP + PhoNLP	Accurate, slow
PhoBERT	Transformer pipeline	High accuracy
ViDeBERTa	Electra-base NER	Balanced
Wikilink	Hyperlink extraction	Bulk ingestion (Typed F1 = 46.9%)

All backends output (name, type) tuples → typed graph labels

Postprocessing: noise filter → org reclassification → deduplication